

Risk Register

#	Risk	Category	Likelihood	Impact	Mitigation
R1	Model fails to reach the 80% minimum accuracy threshold	Technical	High	High	Explore alternative models in Iteration 1. Use cross-validation rather than a single split to get a reliable accuracy estimate before committing to a model.
R2	Confidence score generation cannot be implemented with the current linear regression model	Technical	High	High	Switch to a classification-based model that supports probability output (<code>predict_proba</code>), or build a calibration layer on top of regression output. This must be resolved in Iteration 1 before Iteration 2 can be completed.
R3	Output scale mismatch (0–1 vs 1–10) causes incorrect results in the dashboard	Technical	High	High	Add a rescaling step immediately after the Score Model component. This is a straightforward fix but must be applied before any results are shown to stakeholders or used in aggregated scores.
R4	Spam and manipulated review detection fails to meet the 95% precision / 90% recall requirement	Technical	Medium	High	Design and evaluate a dedicated spam detection module in Iteration 2. Test against labeled examples of spam and manipulated reviews. Track precision and recall explicitly rather than relying on overall accuracy.
R5	Single-word and empty-negative review flagging misses the 95% coverage requirement	Technical	Medium	High	Before the reviews get scored, add a filter step that automatically catches and flags any review that is just one word or has no text at all. Then test that filter to make sure it actually catches them every time.
R6	Persistent output storage is not implemented in time for dashboard development	Technical	Medium	High	Treat output storage as a prerequisite for Iteration 2. Define the output schema early and agree on a storage format (e.g., CSV with timestamps, SQLite, or Azure Blob) before dashboard work begins.
R7	The Streamlit dashboard does not meet the 2-interaction usability requirement	Usability	Medium	Medium	Have a team member pretend to be an end user and test the dashboard by actually clicking through it. Count how many clicks it takes to filter or sort reviews and make sure it never takes more than 2.
R8	Class imbalance in the review dataset biases the model toward extreme ratings	Technical	High	Medium	Before training the model, check if certain ratings show up way more than others in the dataset. If they do, adjust the model so it does not just keep predicting the most common rating. Make sure to test how well it performs on each rating value, not just overall.
R9	Trend tracking and 6-month history retention cannot	Scope	Medium	Medium	Set up the storage system to save every batch of scored reviews with a timestamp from day one. Even if we do not have 6 months of data yet, the system

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	be demonstrated within the project timeline				should be built in a way that shows it is capable of storing and tracking that history over time.
R10	Streamlit is a basic demo tool and was not built to handle 300 users at once. Our dashboard is just a prototype to demonstrate the concept, not a production-ready system	Scope	High	Low	In our documentation, note that the dashboard is just a prototype and not meant for real-world use. Show in the design how it could be scaled up in the future, even if it only runs locally for now.